

PHASE 3 — DAY 1 SCHEMA DESIGN & RECONCILIATION

Project: CMC MIS SIMPLIFIED **Document:** Phase 3 Day 1 Output — Schema Audit + Target Design **Date:** May 16, 2026 **Prepared by:** Claude (AI engineering pair) for Deep Sorathiya (DS) **Inputs received:** O1 ((cmcmis_schema_only.sql) + (cmcmis_table_summary.csv)), O2 (deferred to my analysis), O3 (SA79900, AC77777) **Authority:** Subordinate to (FINAL-DESC-CMC MIS v1.0). Any conflict → FINAL-DESC wins.

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1. EXECUTIVE SUMMARY

Metric	Count
Existing tables in dump	64
Tables with FK constraints	22
Tables with zero rows (likely dead)	5
Backup / orphan tables (cf001-cf004, _bkp, _jun2016)	7
Tables referenced by FK but NOT in dump	1 ((cmms_cont_mst) — vendor master)
Total existing rows across all tables	~390,000+

Metric	Count
Largest table	<code>cmms_jobcard_cal_observations</code> (77,171 rows)
MVP-critical existing tables to KEEP	~22
MVP-critical existing tables to MODIFY	~8
Existing tables to ISOLATE (orphan/dead)	~10
NEW tables to ADD for MVP	~15
Target total table count after Phase 3	~70

Bottom line: The existing database is a real, heavily-used CMMS with ~390K rows of operational data. The **Job Lifecycle** cluster is mature and rich (we will reuse aggressively). The **Identity & RBAC** cluster has a fundamentally wrong shape for our 3-layer RBAC model — we will **NOT modify the existing auth tables**, we will **build a new clean auth stack alongside** and use `cmms_emp_mst` only as the employee directory (lookup source for valid `employee_id`s).

2. CRITICAL FINDINGS (READ FIRST)

FINDING #1 — Plaintext passwords in existing auth

`cmms_userrole_mst.USER_PASSWORD VARCHAR(10)` — passwords are stored in plaintext, max 10 chars. **565 rows**.

Impact: Cannot reuse this table. Violates BR-AUTH (bcrypt required), NFR Security (bcrypt ≥10 rounds). 565 user records will need migration — see Cluster 1 plan below.

FINDING #2 — `cmms_cont_mst` is referenced but missing from dump

4 tables reference `cmms_cont_mst (CMM_CONT_ID)` as FK parent:

- `cmms_amc_mst.AMC_VENDERID`
- `cmms_checklist_mst.CHL_MAKE`
- `cmms equip_mst.EQM_MFRID` ← **CRITICAL** (used by 5,704 equipment records)
- `cmms_inv_mst.INV_MCODE`

Impact: `cmms_cont_mst` is the vendor/manufacturer master. Without its DDL we can't see its columns. **DS must export this table separately.** (See Open Question #1.)

FINDING #3 — Existing RBAC model is incompatible

`cmms_accessright_mst (3221 rows)` uses an **operation-flag matrix**: `(module_id, role_id) → ADD/MODIFY/VIEW/DELETE/PRINT booleans`.

Our locked model (BR-RBAC-03) uses **resource:action permissions**: (`role_id`, `permission_id`) where permission is ('equipment:create'), ('job_card:verify-close'), etc.

These models do NOT map 1:1. We must build new (`permissions`), (`role_permissions`), (`user_roles`) tables. The old (`cmms_accessright_mst`) and (`cmms_module_mst`) are isolated (kept for historical reference, but the runtime never reads from them).

FINDING #4 — 23 existing roles vs locked 5 roles

`cmms_role_mst` has 23 rows. Our locked decision (C1) is **5 roles only**. We need a mapping table from old `role_id` → new `role_id` for the 565 existing user assignments (data migration concern, not schema concern).

FINDING #5 — Composite PK on equipment is unusual

`cmms equip_mst` uses composite PK (`EQM_TYPE`, `EQM_ID`) — 5,704 rows. Type is ('TME') or ('FPE') (likely). Most joins use both columns.

Decision: KEEP composite PK to avoid breaking 5,704 records + 13 dependent tables. New tables we add will use single-column surrogate keys.

FINDING #6 — Orphan backup tables

7 tables look like backups or test artifacts with no FKs and no documentation:

Table	Rows	Likely meaning
<code>cf001</code>	6	Backup of <code>cmms_checklist_hist</code> columns
<code>cf002</code>	553	Backup of <code>cmms_checklist_tasks_hist</code> columns
<code>cf003</code>	570	Backup of <code>cmms_checklist_mst</code> columns
<code>cf004</code>	3,449	Backup of <code>cmms_checklist_tasks</code> columns
<code>chklistvendor</code>	238	Loose junction without FK — <code>chklistno</code> × <code>mfrid</code> × <code>modelno</code>
<code>cmms_parameter_master_bkp</code>	4	Backup of <code>cmms_parameter_master</code>
<code>cmms_parameter_master_jun2016</code>	233	Snapshot from June 2016
<code>cmms_parameter_master_incharge</code>	9	Old/orphan variant

Decision: ISOLATE (rename with `_legacy_` prefix in Phase 2). Not in MVP code path. Not dropped (Constraint #2).

22 jobcard/jobrequest tables with ~280,000+ rows of real history. Schema is verbose (varchar everywhere, no proper status enum), but the relationships are sound. **Strategy: ALTER existing tables to add what we need, do NOT recreate.**

● FINDING #8 — Super Admin IDs format mismatch

DS provided `SA79900` and `AC77777` — both 7 chars, fits `cmms_emp_mst.EMM_ID VARCHAR(7)` ✓.

Concern: Are these existing employee records or new entries to be created? Of the 57 employees in `cmms_emp_mst`, we need to verify these two exist. See Open Question #3.

3. MASTER INVENTORY — All 64 Existing Tables Classified

Legend: ✓ = KEEP as-is · 🛠 = KEEP + MODIFY (ALTER TABLE) · 🗑 = ISOLATE (legacy, not in MVP code) · 🌱 = NEW (will be added) · 🔴 = SECURITY ISSUE

#	Table	Rows	Cluster	Bucket	Action
1	cf001	6	Orphan		Rename (cf001) exclude from
2	cf002	553	Orphan		Rename (cf002) exclude from
3	cf003	570	Orphan		Rename (cf003) exclude from
4	cf004	3,449	Orphan		Rename (cf004) exclude from
5	chklistvendor	238	Orphan		Rename _legacy_c
6	cmms_accessright_mst	3,221	1 Identity		OLD RBAC, i model — nev
7	cmms_amc_mst	0	6 Maintenance / 8 Procurement		Empty, P2 cl only
8	cmms_cal_jobcard_feedback_spec	0	4 Job Lifecycle		Empty + has
9	cmms_checklist_hist	811	4 Job Lifecycle (sub)		Keep for hist
10	cmms_checklist_mst	928	4 Job Lifecycle (sub)		Keep; add au
11	cmms_checklist_tasks	7,536	4 Job Lifecycle (sub)		Junction, ke
12	cmms_checklist_tasks_hist	8,450	4 Job Lifecycle (sub)		Historical, ke
13	cmms_designation_mst	40	2 Organisation		Designations
14	cmms_device_spares_mst	67	6/8 Maint/Procurement		P2 cluster —
15	cmms_division_hist	3,676	3 Equipment		Equipment d history
16	cmms_documentno_mst	151	12 Lookups		Document n sequences

#	Table	Rows	Cluster	Bucket	Action
17	cmms_emp_mst	57	1 Identity		Employee di KEEP, light
18	cmms equip_detail_spec	3	3 Equipment		Has BLOB in empty — P2 of MVP)
19	cmms equip_mst	5,704	3 Equipment		Equipment add new col
20	cmms equip_mst_hist	519	3 Equipment		History, keep
21	cmms equipinst_identification	2,286	3 Equipment		Sub-instrum identification
22	cmms_fault_mst	30	4 Job Lifecycle / 12 Lookups		Fault categor
23	cmms_ins_accuracy_info	1,501	3 Equipment / 5 Calibration		Accuracy/rar instrument
24	cmms_inv_mst	42	8 Procurement		P2 cluster —
25	cmms_jobcard_attendedby_dtl	27,890	4 Job Lifecycle		Engineer ass (junction)
26	cmms_jobcard_awaitinginfo	7,261	4 Job Lifecycle		Awaiting/idle jobcard
27	cmms_jobcard_cal_adjustments_dtl	1,831	4 Job Lifecycle		Calibration a
28	cmms_jobcard_cal_dtl	9,065	4 Job Lifecycle		Calibration d
29	cmms_jobcard_cal_observations	77,171	4 Job Lifecycle		THE observa table)
30	cmms_jobcard_contract_warranty_dtl	17,225	4 / 8		Keep but mo
31	cmms_jobcard_eq_used	38,316	4 Job Lifecycle		Equipment u
32	cmms_jobcard_faulty_category	8,605	4 Job Lifecycle		Fault tag (jur
33	cmms_jobcard_faulty_section	8,131	4 Job Lifecycle		Fault tag (jur
34	cmms_jobcard_insp_maint_dtl	0	4 Job Lifecycle		Empty — de
35	cmms_jobcard_inspection_info	2,214	4 Job Lifecycle		Registration

#	Table	Rows	Cluster	Bucket	Action
36	cmms_jobcard_mst	19,432	4 Job Lifecycle		Core job card cols
37	cmms_jobcard_mst_history	22,143	4 / 10 Audit		History table
38	cmms_jobcard_repair_info	8,118	4 Job Lifecycle		Repair sub-in
39	cmms_jobcard_request_info	19,432	4 Job Lifecycle		Request info
40	cmms_jobcard_request_item_dtl	11,064	4 Job Lifecycle		Request item
41	cmms_jobcard_request_project_dtl	22,316	4 Job Lifecycle		Project link
42	cmms_jobcard_spares equip	2,804	4 / 8		Spares used
43	cmms_jobcard_status_hist	22,214	4 / 10 Audit		THE status h
44	cmms_jobrequest_item_dtl	7,786	4 Job Lifecycle		JR items
45	cmms_jobrequest_mst	21,485	4 Job Lifecycle		Core job req add cols
46	cmms_jobrequest_project_dtl	19,624	4 Job Lifecycle		JR-project li
47	cmms_lineitem_mst	24	8 Procurement		Budget line i
48	cmms_module_mst	163	1 Identity		OLD menu h replaced by p driven UI
49	cmms_parameter_master	337	12 Lookups		THE lookup will moderni
50	cmms_parameter_master_bkp	4	Orphan		Backup — isc
51	cmms_parameter_master_incharge	9	Orphan		Old variant –
52	cmms_parameter_master_jun2016	233	Orphan		Old snapsho
53	cmms_po_mst	115	8 Procurement		P2 cluster, ke
54	cmms_product_mst	32	3 Equipment / 12 Lookups		Instrument t
55	cmms_proj_mst	182	2 Organisation		Project mast
56	cmms_pur_dtl	0	8 Procurement		Empty — P2

#	Table	Rows	Cluster	Bucket	Action
57	cmms_pur_mst	0	8 Procurement		Empty — P2
58	cmms_role_mst	23	1 Identity		23 old roles v new roles
59	cmms_schedule equip_dtl	316	7 Scheduling		P2 — keep
60	cmms_schedule_mst	6	7 Scheduling		P2 — keep
61	cmms_section_mst	293	2 Organisation		Divisions/sec
62	cmms_section_user_mst	294	1 / 2		Old user-sec mapping — r RBAC
63	cmms_task_mst	1,489	4 Job Lifecycle		Task master
64	cmms_userrole_mst	565	1 Identity		Plaintext pa NOT REUSE to new table

Bucket Totals

Bucket	Count	Note
 KEEP as-is	30	Use directly in MVP code
 KEEP + MODIFY	6	Need ALTER TABLE to add new columns
 ISOLATE (legacy)	28	Rename or leave; not in MVP read/write path
 Quarantine	0	(the userrole_mst is isolated and data migrated, not quarantined)
 NEW (to add)	~15	See per-cluster sections below

4. CLUSTER MAPPING — Old Tables → 13 Target Clusters

Cluster 1 — Identity & Access (target 8-10 tables)

Existing tables in this cluster: 6 (all going to ISOLATE)

- cmms_emp_mst — KEEP as employee directory (light ALTER)
- cmms_role_mst — ISOLATE

- `cmms_userrole_mst` — ISOLATE + migrate data
- `cmms_accessright_mst` — ISOLATE
- `cmms_module_mst` — ISOLATE
- `cmms_section_user_mst` — ISOLATE

Target NEW tables (8):

1. `users` 🌱
2. `roles` 🌱
3. `permissions` 🌱
4. `role_permissions` 🌱
5. `user_roles` 🌱
6. `refresh_tokens` 🌱
7. `login_audit` 🌱
8. `password_history` 🌱 *(optional, recommended for defence-grade)*

Total Cluster 1: 1 KEEP + 1 ALTER on `cmms_emp_mst` + 8 NEW = 9 tables in active code path.

Cluster 2 — Organisation (target 4-5 tables)

Existing: ✓ `cmms_section_mst` (293), ✓ `cmms_proj_mst` (182), ✓ `cmms_designation_mst` (40)

No NEW tables needed for MVP. Sketch later: departments, locations.

Cluster 3 — Equipment Master (target 8-10 tables)

Existing (KEEP): `cmms Equip_mst` 🛠️, `cmms Equip_mst_hist` ✓, `cmms Equipinst_identification` ✓, `cmms_ins_accuracy_info` ✓, `cmms_division_hist` ✓, `cmms_product_mst` ✓, `cmms_fault_mst` ✓

ISOLATE: `cmms Equip_detail_spec` (BLOB, almost empty)

NEW:

1. `equipment_status_history` 🌱 (formal state-machine log for PENDING_VERIFICATION → ACTIVE → etc.) *(unless `cmms_division_hist` can be repurposed — see Open Q5)*

Total: 7 existing + 1 new = 8 tables in active code path.

Cluster 4 — Job Lifecycle (target 6-8 tables, but actually 16+ existing)

This is the largest cluster. 22 existing job/jobcard tables. Almost all KEEP.

KEEP (no change): 18 jobcard/jobrequest sub-tables **MODIFY:** `cmms_jobrequest_mst`, `cmms_jobcard_mst` — add new state columns and audit columns **ISOLATE:** `cmms_cal_jobcard_feedback_spec` (0 rows, BLOB), `cmms_jobcard_insp_maint_dtl` (0 rows)

NEW:

1. `job_request_status_history` 🌱 (if `cmms_jobcard_status_hist` can be extended to handle JRs too, this isn't needed — see Open Q6)

Verdict: Cluster 4 is actually fine, the existing schema is rich. We extend, not rebuild.

Cluster 5 — Calibration (target 6-8 tables)

Existing: `cmms_jobcard_cal_dtl`, `cmms_jobcard_cal_observations`, `cmms_jobcard_cal_adjustments_dtl`, `cmms_ins_accuracy_info` — already in jobcard cluster effectively

Strategy: Calibration is currently a child of Job Lifecycle in your schema, not a separate cluster. For MVP we'll keep it that way. P2 may add `calibration_certificates`, `traceability`, `standards`.

Cluster 6 — Maintenance (target 4-5 tables)

Existing: `cmms_amc_mst` (0 rows), `cmms_device_spare_mst` — P2

No MVP work.

Cluster 7 — Scheduling (target 3-4 tables) — P2

`cmms_schedule_mst`, `cmms_schedule equip_dtl` — sketch only.

Cluster 8 — Procurement (target 6-7 tables) — P2

`cmms_po_mst`, `cmms_pur_mst`, `cmms_pur_dtl`, `cmms_inv_mst`, `cmms_lineitem_mst` — sketch only.

Cluster 9 — Documents (target 3-4 tables) — P2 deferred (no file storage in MVP)

Cluster 10 — Audit & Logs (target 3-4 tables)

Existing: `cmms_jobcard_status_hist` (✅, 22,214 rows), `cmms_jobcard_mst_history` (✅,

22,143 rows), `cmms_equip_mst_hist` (✅, 519), `cmms_division_hist` (✅, 3,676)

These are ENTITY-SPECIFIC history tables — useful but not enough.

NEW:

- 1. `audit_log` 🌱 — generic write-audit log (BR-AUD-01)
- 2. `audit_log_changes` 🌱 — JSON diff per entry (BR-AUD-02)
- 3. `export_audit` 🌱 — BR-AUD-03

Total: 4 existing + 3 new = 7 in active path.

Cluster 11 — Notifications (target 3 tables) — P2 (no email in MVP)

Cluster 12 — Lookups (target 5-7 tables)

Existing: `cmms_parameter_master` 🔑 (337, primary lookup), `cmms_documentno_mst` ✅, `cmms_product_mst` ✅, `cmms_fault_mst` ✅, `cmms_task_mst` ✅, `cmms_designation_mst` ✅

ISOLATE: `cmms_parameter_master_bkp`, `cmms_parameter_master_incharge`, `cmms_parameter_master_jun2016`

Strategy: `cmms_parameter_master` already has shape `((CategoryID, CategoryDescription, SrID, Value))` which is a perfectly serviceable key-value lookup. We add a PK + audit columns.

No NEW tables needed for MVP.

Cluster 13 — Reporting — P2

5. RECONCILIATION MATRIX — SUMMARY



NEW MVP TABLES (~12 in MVP-critical clusters)

Cluster 1 (Identity) : 8 new (users, roles, permissions, etc.)
Cluster 3 (Equipment) : 1 new (equipment_status_history)
Cluster 4 (Job) : 0-1 new
Cluster 10 (Audit) : 3 new (audit_log, audit_log_changes, export_audit)

=

ACTIVE MVP SCHEMA = ~36 existing + ~12 new = ~48 tables
Plus ~28 legacy tables left untouched but renamed → ~76 total
✓ Aligned with FINAL-DESC target "~70 tables"

6. CLUSTER 1 — Identity & Access (AUTH) — FULL DEEP DIVE

This is the most important cluster for MVP. The existing auth is **fundamentally incompatible** with our locked 3-layer RBAC model + bcrypt + JWT. We do NOT modify existing tables; we build a clean new stack and use `cmms_emp_mst` as a read-only employee directory.

6.1 Strategic Decision Diagram

EXISTING (DO NOT BREAK)

`cmms_emp_mst` (57 rows)
employee directory
(kept as source of truth
for valid `employee_ids`)

← FK →

NEW (MVP RUNTIME READS THESE)

`users` (NEW)
auth identity
(bcrypt password + flags)
|
| one role per user
▼
`user_roles` (NEW)
(PK on `user_id` enforces
BR-RBAC-02: ONE role)
|
▼
`roles` (NEW — exactly 5 rows)
|
| many-to-many
▼
`role_permissions` (NEW)
|
▼

permissions (NEW – ~50 rows)
(resource:action)

ISOLATED (rename _legacy_*)

- cmms_role_mst (23 roles, archived)
- cmms_userrole_mst (565 user-role-pwd rows, MIGRATED then archived)
- cmms_accessright_mst (3221 module-role flags, archived)
- cmms_module_mst (163 menu items, archived)
- cmms_section_user_mst (294 mappings, archived)

6.2 Data Migration Plan (existing 565 user-role rows → new tables)

Source: `cmms_userrole_mst` (565 rows). **Target:** `users` + `user_roles`.

Old column	Old type	New target	New type	Notes
<code>USER_ID</code>	varchar(7)	<code>users.employee_id</code>	varchar(7)	Direct copy
<code>USER_PASSWORD</code>	varchar(10)	<code>users.password_hash</code>	varchar(60)	CANNOT migrate directly bcrypt the plaintext on first login OR force password reset for all (recommended)
<code>USER_STATE</code>	tinyint(1)	<code>users.is_active</code>	tinyint(1)	Direct
<code>USER_DIVISION_ID</code>	int	(dropped)	—	Division is now in <code>cmms_section_user_mst</code> or stored in <code>cmms_emp_mst.EMM_DEPT</code>
<code>USER_ROLE</code>	int (1..23)	<code>user_roles.role_id</code>	tinyint (1..5)	Requires role-mapping table — see Open Q4

Recommended migration approach (defence-grade):

1. Seed `users` row for each `cmms_userrole_mst.USER_ID` with `password_hash = NULL` and `password_must_change = TRUE`.
2. Force password reset on first login for every existing user.
3. Do NOT auto-migrate plaintext passwords — even if bcrypted, the original was 10 chars and probably weak.

6.3 Complete DDL — Cluster 1 (Identity & Access)

sql

```

-- =====
-- CLUSTER 1: IDENTITY & ACCESS
-- Locked decisions: D11, D12, D17, D18, BR-AUTH-*, BR-RBAC-*
-- =====

-- 1.0 – ALTER existing employee directory (KEEP + light modify)
-- =====
-- We do NOT touch business columns. We add a deterministic activity
-- index so auth can fast-filter to "loginable" employees.

ALTER TABLE `cmms_emp_mst`
  ADD INDEX `idx_emm_active` (`EMM_INACTIVE`);

-- (Note: we intentionally do NOT add a password column to cmms_emp_mst.
-- Auth identity is a separate concern from employee record. New
-- `users` table owns auth state. This is industry-standard separation.)

-- 1.1 – USERS (NEW)
-- =====
-- The auth-aware user record. ONE row per loginable employee.
-- Bound to cmms_emp_mst (employee directory) via FK on employee_id.

CREATE TABLE `users` (
  `user_id`          BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,
  `employee_id`      VARCHAR(7)      NOT NULL,
  `password_hash`    VARCHAR(60)     NULL DEFAULT NULL
                      COMMENT 'bcrypt hash, 60 chars. NULL until first set.',
  `password_must_change` TINYINT(1)  NOT NULL DEFAULT 1
                      COMMENT 'TRUE on bootstrap or admin reset',
  `password_changed_at` DATETIME(6)   NULL DEFAULT NULL,
  `is_active`        TINYINT(1)      NOT NULL DEFAULT 1
                      COMMENT 'BR-AUTH-07: deactivated cannot login',
  `is_locked`        TINYINT(1)      NOT NULL DEFAULT 0
                      COMMENT 'Auto-lock after N failed attempts',
  `failed_login_count` SMALLINT UNSIGNED NOT NULL DEFAULT 0,
  `last_login_at`    DATETIME(6)     NULL DEFAULT NULL,
  `last_login_ip`    VARCHAR(45)     NULL DEFAULT NULL
                      COMMENT 'IPv6-ready',
  `created_at`       DATETIME(6)     NOT NULL DEFAULT CURRENT_TIMESTAMP(6),
  `created_by`       VARCHAR(7)      NULL DEFAULT NULL,
  `updated_at`       DATETIME(6)     NOT NULL DEFAULT CURRENT_TIMESTAMP(6)
                      ON UPDATE CURRENT_TIMESTAMP(6),
  `updated_by`       VARCHAR(7)      NULL DEFAULT NULL,
  PRIMARY KEY (`user_id`),
  UNIQUE KEY `uk_users_employee_id` (`employee_id`),

```

```

CONSTRAINT `fk_users_employee`
  FOREIGN KEY (`employee_id`)
    REFERENCES `cmms_emp_mst` (`EMM_ID`),
INDEX `idx_users_active` (`is_active`, `is_locked`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

-- 1.2 – ROLES (NEW – exactly 5 system roles per C1)

```

CREATE TABLE `roles` (
  `role_id`          TINYINT UNSIGNED NOT NULL,
  `role_code`        VARCHAR(30)      NOT NULL
                        COMMENT 'SUPER_ADMIN | LAB_IN_CHARGE | LAB_ENGINEER | NORMAL_US
  `role_name`        VARCHAR(60)      NOT NULL,
  `role_description` VARCHAR(255)      NULL DEFAULT NULL,
  `is_system`        TINYINT(1)       NOT NULL DEFAULT 1
                        COMMENT 'System roles cannot be deleted via UI',
  `created_at`       DATETIME(6)      NOT NULL DEFAULT CURRENT_TIMESTAMP(6),
  PRIMARY KEY (`role_id`),
  UNIQUE KEY `uk_roles_code` (`role_code`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

-- Seed the 5 locked roles (run from migration, not app code)

```

INSERT INTO `roles` (`role_id`, `role_code`, `role_name`, `role_description`) VALUES
(1, 'SUPER_ADMIN',    'Super Admin',      'Master data + RBAC + system integrity'),
(2, 'LAB_IN_CHARGE',  'Lab In-Charge',    'Approve, assign, verify, close'),
(3, 'LAB_ENGINEER',   'Lab Engineer',     'Execute jobs, fill cards, observations'),
(4, 'NORMAL_USER',    'Normal User',      'Raise requests, register equipment'),
(5, 'VIEW_ONLY',      'View-Only',        'Read-only auditor / management oversight')

```

-- 1.3 – PERMISSIONS (NEW – atomic resource:action)

```

CREATE TABLE `permissions` (
  `permission_id`    SMALLINT UNSIGNED NOT NULL AUTO_INCREMENT,
  `permission_code`  VARCHAR(80)        NOT NULL
                        COMMENT 'e.g., equipment:create | job_card:verify-close',
  `resource`         VARCHAR(40)        NOT NULL
                        COMMENT 'e.g., equipment, job_card, audit_log',
  `action`           VARCHAR(40)        NOT NULL
                        COMMENT 'e.g., create, read-list, verify, reopen',
  `description`      VARCHAR(255)       NULL DEFAULT NULL,
  `is_system`        TINYINT(1)         NOT NULL DEFAULT 1,
  `created_at`       DATETIME(6)        NOT NULL DEFAULT CURRENT_TIMESTAMP(6),
  PRIMARY KEY (`permission_id`),

```

```

    UNIQUE KEY `uk_perm_code` (`permission_code`),
    INDEX `idx_perm_resource` (`resource`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

-- Seed permissions per FINAL-DESC §6 (run from migration)
-- Subset shown – full list mirrors the permission matrix exactly
INSERT INTO `permissions` (`permission_code`, `resource`, `action`, `description`) VALUES
-- Auth & Identity
('auth:login', 'auth', 'login', 'Submit credentials'),
('auth:logout', 'auth', 'logout', 'End session'),
('auth:refresh-token', 'auth', 'refresh-token', 'Refresh JWT token'),
('me:read', 'me', 'read', 'Read own profile'),
-- User mgmt
('user:read-list', 'user', 'read-list', 'List users'),
('user:role-assign', 'user', 'role-assign', 'Assign/change role'),
('user:activate', 'user', 'activate', 'Activate user'),
('user:deactivate', 'user', 'deactivate', 'Deactivate user'),
-- Equipment
('equipment:read-list', 'equipment', 'read-list', 'List equipment'),
('equipment:read-detail', 'equipment', 'read-detail', 'View equipment details'),
('equipment:create', 'equipment', 'create', 'Register new equipment'),
('equipment:update', 'equipment', 'update', 'Edit equipment'),
('equipment:verify', 'equipment', 'verify', 'PENDING_VERIFICATION'),
('equipment:condemn', 'equipment', 'condemn', 'Mark CONDEMNED'),
('equipment:delete', 'equipment', 'delete', 'Hard delete'),
-- Job Requests
('job_request:create', 'job_request', 'create', 'Create JR'),
('job_request:read-own', 'job_request', 'read-own', 'Read own JR'),
('job_request:read-all', 'job_request', 'read-all', 'Read all JR'),
('job_request:approve', 'job_request', 'approve', 'Approve JR'),
('job_request:reject', 'job_request', 'reject', 'Reject JR with reason'),
('job_request:assign-engineer', 'job_request', 'assign-engineer', 'Assign LE to JR'),
-- Job Cards
('job_card:read-list', 'job_card', 'read-list', 'List JCs'),
('job_card:read-detail', 'job_card', 'read-detail', 'View JC details'),
('job_card:start-work', 'job_card', 'start-work', 'Mark IN-PROGRESS'),
('job_card:update-tasks', 'job_card', 'update-tasks', 'Update task list'),
('job_card:complete', 'job_card', 'complete', 'Mark COMPLETE'),
('job_card:verify-close', 'job_card', 'verify-close', 'Verify + close'),
('job_card:reopen', 'job_card', 'reopen', 'Reopen closed JC'),
('job_card:generate-pdf', 'job_card', 'generate-pdf', 'Download JC PDF'),
-- Dashboard & Inquiry
('dashboard:view', 'dashboard', 'view', 'Dashboard'),
('inquiry:search-vendors', 'inquiry', 'search-vendors', 'Vendor search'),
('inquiry:search-products', 'inquiry', 'search-products', 'Product search'),
('inquiry:search-job-cards', 'inquiry', 'search-job-cards', 'JC search'),
('inquiry:search-instruments', 'inquiry', 'search-instruments', 'Instrument search')

```



```

-- Master Data (Phase 2)
('master:employees:manage', 'master', 'employees:manage', 'Mgmt master
('master:vendors:manage', 'master', 'vendors:manage', 'Mgmt master
('master:equipment-types:manage', 'master', 'equipment-types:manage', 'Mgmt eq ty
('master:divisions:manage', 'master', 'divisions:manage', 'Mgmt divisi
('master:lookup-values:manage', 'master', 'lookup-values:manage', 'Mgmt lookup
-- Audit
('audit_log:read', 'audit_log', 'read', 'Read audit
('export:trigger', 'export', 'trigger', 'Trigger exp

-- 1.4 – ROLE_PERMISSIONS (NEW – many-to-many junction)
-- -----

CREATE TABLE `role_permissions` (
  `role_id` TINYINT UNSIGNED NOT NULL,
  `permission_id` SMALLINT UNSIGNED NOT NULL,
  `granted_at` DATETIME(6) NOT NULL DEFAULT CURRENT_TIMESTAMP(6),
  `granted_by` VARCHAR(7) NULL DEFAULT NULL,
  PRIMARY KEY (`role_id`, `permission_id`),
  CONSTRAINT `fk_rp_role`
    FOREIGN KEY (`role_id`) REFERENCES `roles` (`role_id`) ON DELETE CASCADE,
  CONSTRAINT `fk_rp_permission`
    FOREIGN KEY (`permission_id`) REFERENCES `permissions` (`permission_id`) ON DELETE CASCADE,
  INDEX `idx_rp_perm` (`permission_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

-- Seed role-permission grants per FINAL-DESC §6 permission matrix
-- (Generated by mapping the ✓/× matrix into rows. Below is a sample;
-- full grants live in seed/03_role_permissions.sql.)
-- Super Admin gets ALL permissions:
INSERT INTO `role_permissions` (`role_id`, `permission_id`)
SELECT 1, `permission_id` FROM `permissions`;
-- Other role grants are explicit per the matrix – see seed file.

-- 1.5 – USER_ROLES (NEW – BR-RBAC-02: exactly ONE role per user)
-- -----

-- The PK on `user_id` (not composite) enforces "one role per user".

CREATE TABLE `user_roles` (
  `user_id` BIGINT UNSIGNED NOT NULL,
  `role_id` TINYINT UNSIGNED NOT NULL,
  `assigned_at` DATETIME(6) NOT NULL DEFAULT CURRENT_TIMESTAMP(6),
  `assigned_by` VARCHAR(7) NULL DEFAULT NULL,
  PRIMARY KEY (`user_id`), -- BR-RBAC-02 enforced here
  CONSTRAINT `fk_ur_user`

```

```

        FOREIGN KEY (`user_id`) REFERENCES `users` (`user_id`) ON DELETE CASCADE,
    CONSTRAINT `fk_ur_role`
        FOREIGN KEY (`role_id`) REFERENCES `roles` (`role_id`),
    INDEX `idx_ur_role` (`role_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

-- 1.6 – REFRESH_TOKENS (NEW – D17: 7-day refresh)

```

CREATE TABLE `refresh_tokens` (
    `token_id`          BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,
    `user_id`           BIGINT UNSIGNED NOT NULL,
    `token_hash`        VARCHAR(255)     NOT NULL
                        COMMENT 'SHA-256 hash of raw token; raw token never stored',
    `issued_at`         DATETIME(6)      NOT NULL DEFAULT CURRENT_TIMESTAMP(6),
    `expires_at`        DATETIME(6)      NOT NULL,
    `revoked_at`        DATETIME(6)      NULL DEFAULT NULL,
    `revoked_reason`    VARCHAR(60)      NULL DEFAULT NULL
                        COMMENT 'logout | rotated | admin_revoke | password_change',
    `user_agent`        VARCHAR(500)     NULL DEFAULT NULL,
    `ip_address`        VARCHAR(45)      NULL DEFAULT NULL,
    PRIMARY KEY (`token_id`),
    UNIQUE KEY `uk_rt_hash` (`token_hash`),
    CONSTRAINT `fk_rt_user`
        FOREIGN KEY (`user_id`) REFERENCES `users` (`user_id`) ON DELETE CASCADE,
    INDEX `idx_rt_user_expires` (`user_id`, `expires_at`),
    INDEX `idx_rt_expires` (`expires_at`)
                        COMMENT 'For background cleanup of expired tokens'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

-- 1.7 – LOGIN_AUDIT (NEW – BR-AUTH-06: every login attempt logged)

```

CREATE TABLE `login_audit` (
    `audit_id`          BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,
    `employee_id`       VARCHAR(7)      NOT NULL
                        COMMENT 'Not user_id – failed logins may have no matching user',
    `attempt_at`        DATETIME(6)      NOT NULL DEFAULT CURRENT_TIMESTAMP(6),
    `outcome`           ENUM(
        'SUCCESS',
        'FAILED_BAD_PASSWORD',
        'FAILED_LOCKED',
        'FAILED_INACTIVE',
        'FAILED_NOT_FOUND',
        'FAILED_MUST_CHANGE_PW',
    )
)

```

```

        'LOGOUT'
    ) NOT NULL,
    `ip_address` VARCHAR(45) NULL DEFAULT NULL,
    `user_agent` VARCHAR(500) NULL DEFAULT NULL,
    `notes` VARCHAR(255) NULL DEFAULT NULL,
    PRIMARY KEY (`audit_id`),
    INDEX `idx_la_emp_time` (`employee_id`, `attempt_at`),
    INDEX `idx_la_time` (`attempt_at`),
    INDEX `idx_la_outcome` (`outcome`, `attempt_at`)
    COMMENT 'For "failed logins last 24h" dashboards'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

```

```

-- 1.8 – PASSWORD_HISTORY (NEW – defence-grade nicety, OPTIONAL but recommended)
-- -----
-- Prevents reuse of last N passwords. ISRO SAC-like environment usually
-- mandates this. Mark as OPTIONAL – confirm with DS before keeping.

```

```

CREATE TABLE `password_history` (
    `history_id` BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,
    `user_id` BIGINT UNSIGNED NOT NULL,
    `password_hash` VARCHAR(60) NOT NULL,
    `changed_at` DATETIME(6) NOT NULL DEFAULT CURRENT_TIMESTAMP(6),
    `changed_by` VARCHAR(7) NULL DEFAULT NULL
    COMMENT 'NULL if self-change, else admin who reset it',
    PRIMARY KEY (`history_id`),
    CONSTRAINT `fk_ph_user`
    FOREIGN KEY (`user_id`) REFERENCES `users` (`user_id`) ON DELETE CASCADE,
    INDEX `idx_ph_user_time` (`user_id`, `changed_at` DESC)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;

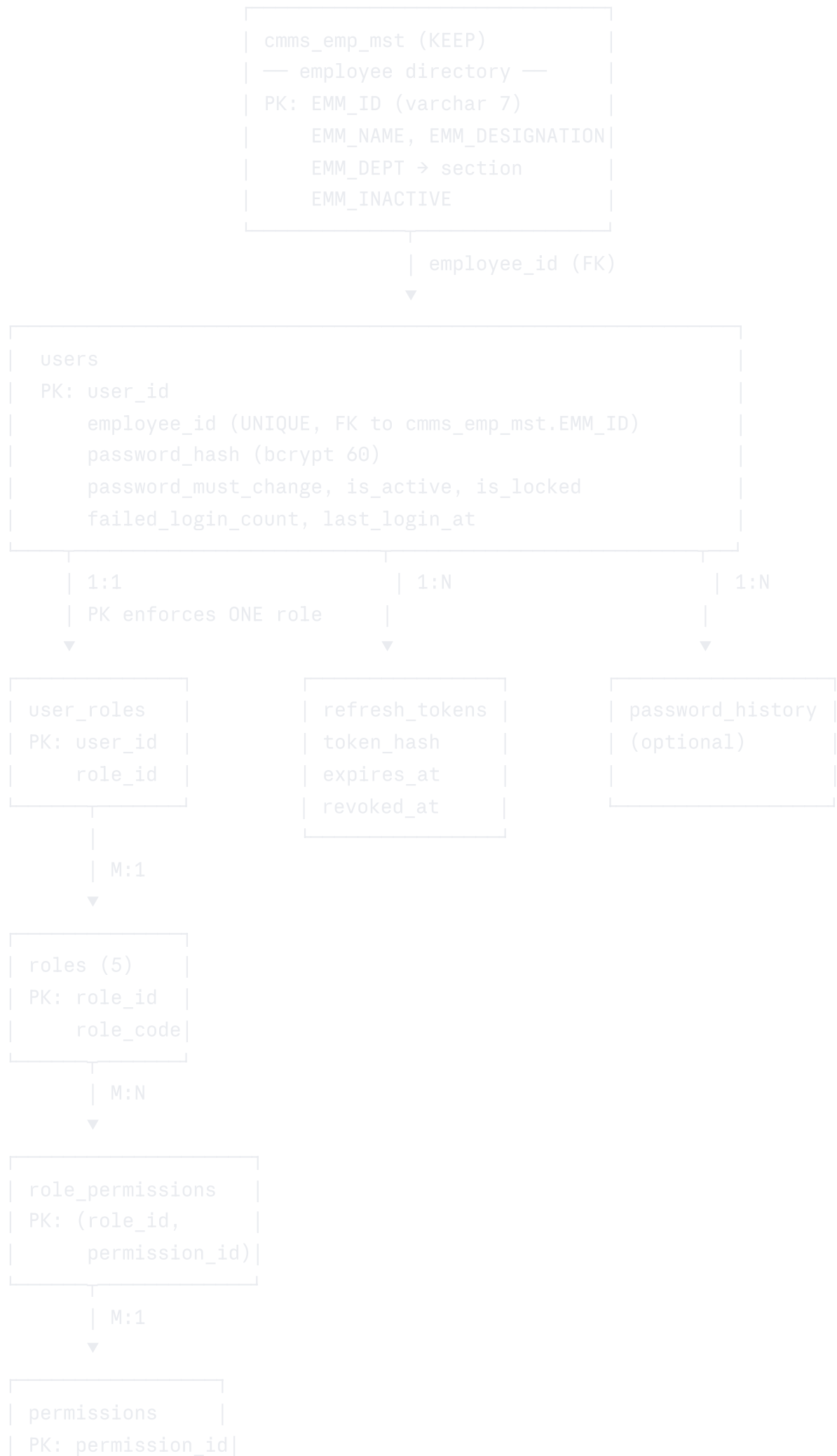
```

6.4 Old vs New — Auth Comparison Table

Concern	Existing (cmms_*)	New (clean stack)	Why we change
Password storage	<u>USER_PASSWORD varchar(10)</u> plaintext	<u>users.password_hash</u> <u>varchar(60)</u> bcrypt	● Plaintext is un bcrypt locked in Security + S5.
User identity	Composite (USER_ID, USER_DIVISION_ID)	Single PK <u>user_id BIGINT</u> + UNIQUE <u>employee_id</u>	Composite PK fo redundant rows i moves division. I model: one user, Division lives in <u>cmms_emp_mst</u>

Concern	Existing (cmms_*)	New (clean stack)	Why we change
Role count	23 (in <code>cmms_role_mst</code>)	5 (in <code>roles</code>)	Locked C1: 5 role 23 will be archived existing user assignments need a role-mapping Open Q4).
Roles per user	Could have multiple via (USER_ID, DIVISION) pairs	Exactly 1 (PK on <code>user_roles.user_id</code>)	Locked BR-RBA Enforced at the system — not just business
Permission model	<code>(MOD_ID, ROLE_ID) → 5 boolean flags</code> (ADD/MODIFY/VIEW/DELETE/PRINT)	<code>(role_id, permission_id)</code> where permission is <code>resource:action</code>	Locked BR-RBA never checks role model couldn't be <code>job_card:version</code> distinct from <code>job_card:comment</code>
Token storage	None (session-based, no JWT)	<code>refresh_tokens</code> table with SHA-256 hash	Locked D17: JWT refresh cookie 70 not-token at rest standard.
Login audit	None visible	<code>login_audit</code> with outcome enum	Locked BR-AUTH coverage of login
Inactive user	<code>USER_STATE tinyint(1)</code>	<code>users.is_active</code> + <code>users.is_locked</code> separate	Inactive = admin deactivated. Locked locked from failed Different states, and recovery paths.
Password rotation	Not enforced	<code>password_must_change</code> flag + optional <code>password_history</code>	Bootstrap forces <code>password_history</code> grade nicety.
Failed-attempt tracking	None	<code>users.failed_login_count</code> + <code>login_audit</code>	Defence-grade. I attempts.
Employee directory	Tangled with auth in <code>cmms_userrole_mst</code>	Separated: <code>cmms_emp_mst</code> = directory (KEEP), <code>users</code> = auth identity	Clean separation Employee data li business already

6.5 Auth Module ER Diagram



```
| resource |  
| action   |  
|-----|
```

```
| login_audit (write-only by auth layer) |  
| employee_id, outcome, attempt_at, ip   |
```

6.6 Bootstrap Seed — Super Admin (per D11, D18, BR-AUTH-05)

```
sql  
  
-- Bootstrap migration: runs once on first deploy.  
-- Reads SUPER_ADMIN_EMPLOYEE_IDS env var = "SA79900,AC77777"  
-- DS provided: SA79900, AC77777  
  
-- Step 1: Verify both employee IDs exist in cmms_emp_mst  
-- (If not, migration fails with clear error – DS adds employees first.)  
  
-- Step 2: Create users rows  
INSERT INTO `users`  
  (`employee_id`, `password_hash`, `password_must_change`, `is_active`, `created_by`  
VALUES  
  ('SA79900', NULL, 1, 1, 'BOOTSTRAP'),    -- password set on first login  
  ('AC77777', NULL, 1, 1, 'BOOTSTRAP');  
  
-- Step 3: Grant SUPER_ADMIN role to both  
INSERT INTO `user_roles` (`user_id`, `role_id`, `assigned_by`)  
SELECT u.user_id, 1, 'BOOTSTRAP'  
FROM `users` u  
WHERE u.employee_id IN ('SA79900', 'AC77777');  
  
-- Step 4: Record in audit  
INSERT INTO `audit_log`  
  (`actor_employee_id`, `action`, `entity_type`, `entity_id`, `notes`)  
VALUES  
  ('BOOTSTRAP', 'USER_CREATE', 'user', 'SA79900', 'Super Admin seed via env CSV'),  
  ('BOOTSTRAP', 'USER_CREATE', 'user', 'AC77777', 'Super Admin seed via env CSV'),  
  ('BOOTSTRAP', 'ROLE_ASSIGN', 'user', 'SA79900', 'role=SUPER_ADMIN'),  
  ('BOOTSTRAP', 'ROLE_ASSIGN', 'user', 'AC77777', 'role=SUPER_ADMIN');
```

7. CLUSTER 3 — Equipment Master — Reconciliation + DDL

7.1 Strategy: KEEP + ALTER (don't recreate)

`cmms_equip_mst` has 5,704 rows + 13 dependent tables. Recreating is reckless. We **ALTER** to add what's missing for our locked rules:

What's missing for our locked rules:

- BR-EQP-09: `registered_by` + `verified_by` + timestamps → `EQM_CREATED_BY` exists ✓
but no `verified_by`, `verified_on`
- BR-EQP-10: PENDING_VERIFICATION default → status column exists (`EQM_DIV_STATUS`)
but no formal enum
- D10: New equipment defaults to PENDING_VERIFICATION → not enforced

7.2 DDL — Cluster 3 ALTERs and NEW

```
sql
```

```
-- CLUSTER 3: EQUIPMENT MASTER
```

```
-- 3.1 ALTER cmms equip_mst – add what's missing for BR-EQP rules
```

```
ALTER TABLE `cmms_equip_mst`
-- BR-EQP-09: verification fields
ADD COLUMN `EQM_VERIFIED_BY` VARCHAR(7) NULL DEFAULT NULL AFTER `EQM_CREATED_BY`,
ADD COLUMN `EQM_VERIFIED_ON` DATETIME(6) NULL DEFAULT NULL AFTER `EQM_VERIFIED_BY`,
-- D10 / BR-EQP-10: formal MVP status column
ADD COLUMN `EQM_MVP_STATUS` ENUM(
    'PENDING_VERIFICATION',
    'ACTIVE',
    'UNDER_CALIBRATION',
    'UNDER_REPAIR',
    'OUT_OF_TOLERANCE',
    'QUARANTINED',
    'CONDEMNED',
    'RETIRED'
) NOT NULL DEFAULT 'PENDING_VERIFICATION' AFTER `EQM_DIV_STATUS`,
ADD COLUMN `EQM_MVP_STATUS_AT` DATETIME(6) NULL DEFAULT NULL AFTER `EQM_MVP_STATUS`,
ADD INDEX `idx_equip_mvp_status` (`EQM_MVP_STATUS`),
ADD INDEX `idx_equip_cal_due` (`EQM_CAL_DUE_DATE`),
ADD INDEX `idx_equip_div` (`EQM_DIVID`);
```

```
-- Note: existing 5,704 rows will get NULL EQM_VERIFIED_BY; per business
-- decision, we backfill all existing as 'ACTIVE' (data migration).
```

```
UPDATE `cmms_equip_mst`
SET `EQM_MVP_STATUS` = 'ACTIVE',
    `EQM_MVP_STATUS_AT` = COALESCE(`EQM_UPDATED_ON`, `EQM_CREATED_ON`)
WHERE `EQM_MVP_STATUS` = 'PENDING_VERIFICATION';
```

```
-- 3.2 NEW: equipment_status_history (formal state-machine log)
```

```
-- BR-EQP-06: state transitions logged. `cmms_division_hist` tracks
-- division/status but not the full state machine, so we add this.
```

```
CREATE TABLE `equipment_status_history` (
    `history_id` BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,
    `eqm_type` VARCHAR(15) NOT NULL,
    `eqm_id` INT NOT NULL,
    `from_status` VARCHAR(30) NULL DEFAULT NULL
    COMMENT 'NULL on initial create (= PENDING_VERIFICATION)',
    `to_status` VARCHAR(30) NOT NULL,
```



```
`transitioned_at`    DATETIME(6)      NOT NULL DEFAULT CURRENT_TIMESTAMP(6),
`transitioned_by`    VARCHAR(7)       NOT NULL,
`reason`             VARCHAR(500)    NULL DEFAULT NULL,
`related_job_card`   VARCHAR(9)     NULL DEFAULT NULL
                    COMMENT 'If transition was triggered by a job card',
PRIMARY KEY (`history_id`),
CONSTRAINT `fk_esh_equip`
  FOREIGN KEY (`eqm_type`, `eqm_id`)
  REFERENCES `cmms_equip_mst` (`EQM_TYPE`, `EQM_ID`),
INDEX `idx_esh_equip_time` (`eqm_type`, `eqm_id`, `transitioned_at` DESC)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

8. CLUSTER 4 — Job Lifecycle — Reconciliation + DDL

8.1 Strategy: ALTER existing, almost no NEW tables

Job Lifecycle is the most data-rich cluster. 22 existing tables, ~280K rows, mature relationships. We extend.

8.2 DDL — Cluster 4

sql

-- CLUSTER 4: JOB LIFECYCLE

-- 4.1 ALTER cmms_jobrequest_mst – add formal MVP status enum

```
ALTER TABLE `cmms_jobrequest_mst`
  ADD COLUMN `JR_MVP_STATUS` ENUM(
    'DRAFT',
    'SUBMITTED',
    'ASSIGNED',
    'IN_PROGRESS',
    'COMPLETED',
    'VERIFIED_CLOSED',
    'REJECTED',
    'REOPENED'
  ) NOT NULL DEFAULT 'DRAFT' AFTER `JR_REQUEST_TYPE`,
  ADD COLUMN `JR_MVP_STATUS_AT` DATETIME(6) NULL DEFAULT NULL,
  ADD COLUMN `JR_APPROVED_BY` VARCHAR(7) NULL DEFAULT NULL,
  ADD COLUMN `JR_APPROVED_ON` DATETIME(6) NULL DEFAULT NULL,
  ADD COLUMN `JR_REJECTED_BY` VARCHAR(7) NULL DEFAULT NULL,
  ADD COLUMN `JR_REJECTED_ON` DATETIME(6) NULL DEFAULT NULL,
  ADD COLUMN `JR_REJECTION_REASON` VARCHAR(500) NULL DEFAULT NULL,
  ADD COLUMN `JR_PRIORITY` ENUM('LOW', 'NORMAL', 'HIGH', 'URGENT') NOT NULL DEFAULT 'NO',
  ADD INDEX `idx_jr_status` (`JR_MVP_STATUS`),
  ADD INDEX `idx_jr_submittedby` (`JR_SUBMITTEDBYID`),
  ADD INDEX `idx_jr_division` (`JR_DIVISION`);
```

-- Backfill existing 21,485 rows: derive MVP status from existing data

```
UPDATE `cmms_jobrequest_mst`
  SET `JR_MVP_STATUS` = CASE
    WHEN `JR_SECTIONJOB_NO` IS NOT NULL THEN 'ASSIGNED'
    ELSE 'SUBMITTED'
  END
WHERE `JR_MVP_STATUS` = 'DRAFT';
```

-- 4.2 ALTER cmms_jobcard_mst – add formal MVP status enum + audit

```
ALTER TABLE `cmms_jobcard_mst`
  ADD COLUMN `JM_MVP_STATUS` ENUM(
    'ASSIGNED',
    'IN_PROGRESS',
    'COMPLETED',
```

```
'VERIFIED_CLOSED',
'REOPENED'
) NOT NULL DEFAULT 'ASSIGNED' AFTER `JM_JobStatus`,
ADD COLUMN `JM_VERIFIED_BY` VARCHAR(7) NULL DEFAULT NULL,
ADD COLUMN `JM_VERIFIED_ON` DATETIME(6) NULL DEFAULT NULL,
ADD COLUMN `JM_REOPENED_REASON` VARCHAR(500) NULL DEFAULT NULL,
ADD INDEX `idx_jc_status` (`JM_MVP_STATUS`),
ADD INDEX `idx_jc equip` (`JM_EQM_TYPE`, `JM_EQM_ID`),
ADD INDEX `idx_jc_recd_date` (`JM_JCRecdDate`);

-- (cmms_jobcard_status_hist already exists with 22,214 rows – KEEP as the
-- formal state-history table. We just write our new MVP status into it
-- going forward.)
```

9. CLUSTER 10 — Audit & Logs — Fresh Design

9.1 Strategy: existing history tables are entity-specific; we add a generic write-audit log

```
sql
```

```
-- CLUSTER 10: AUDIT & LOGS
```

```
-- 10.1 audit_log (NEW – BR-AUD-01: every write logged)
```

```
CREATE TABLE `audit_log` (  
  `audit_id` BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,  
  `actor_employee_id` VARCHAR(7) NOT NULL  
    COMMENT 'EMM_ID of acting user, or BOOTSTRAP/SYSTEM',  
  `actor_role_code` VARCHAR(30) NULL DEFAULT NULL,  
  `action` VARCHAR(60) NOT NULL  
    COMMENT 'EQUIPMENT_CREATE | JOB_REQUEST_APPROVE | etc.',  
  `entity_type` VARCHAR(40) NOT NULL  
    COMMENT 'equipment | job_request | job_card | user',  
  `entity_id` VARCHAR(50) NOT NULL  
    COMMENT 'PK of the affected entity (stringified)',  
  `occurred_at` DATETIME(6) NOT NULL DEFAULT CURRENT_TIMESTAMP(6),  
  `ip_address` VARCHAR(45) NULL DEFAULT NULL,  
  `user_agent` VARCHAR(500) NULL DEFAULT NULL,  
  `request_id` VARCHAR(40) NULL DEFAULT NULL  
    COMMENT 'Correlates with pino-http log line',  
  `notes` VARCHAR(500) NULL DEFAULT NULL,  
  PRIMARY KEY (`audit_id`),  
  INDEX `idx_al_entity` (`entity_type`, `entity_id`, `occurred_at` DESC),  
  INDEX `idx_al_actor` (`actor_employee_id`, `occurred_at` DESC),  
  INDEX `idx_al_action` (`action`, `occurred_at` DESC),  
  INDEX `idx_al_time` (`occurred_at` DESC)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

```
-- 10.2 audit_log_changes (NEW – BR-AUD-02: before/after JSON diff)
```

```
CREATE TABLE `audit_log_changes` (  
  `change_id` BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,  
  `audit_id` BIGINT UNSIGNED NOT NULL,  
  `field_name` VARCHAR(80) NOT NULL,  
  `before_value` TEXT NULL DEFAULT NULL,  
  `after_value` TEXT NULL DEFAULT NULL,  
  PRIMARY KEY (`change_id`),  
  CONSTRAINT `fk_alc_audit`  
    FOREIGN KEY (`audit_id`) REFERENCES `audit_log` (`audit_id`) ON DELETE CASCADE,  
  INDEX `idx_alc_audit` (`audit_id`)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

```
-- 10.3 export_audit (NEW – BR-AUD-03: exports logged with user + IDs)
-- =====

CREATE TABLE `export_audit` (
  `export_id`          BIGINT UNSIGNED NOT NULL AUTO_INCREMENT,
  `actor_employee_id`  VARCHAR(7)      NOT NULL,
  `export_type`        VARCHAR(40)     NOT NULL
                        COMMENT 'JOB_CARD_PDF | CAL_CERT_PDF | future EXCEL_*',
  `record_ids`        TEXT             NOT NULL
                        COMMENT 'Comma-separated or JSON array of PK(s)',
  `occurred_at`        DATETIME(6)     NOT NULL DEFAULT CURRENT_TIMESTAMP(6),
  `ip_address`         VARCHAR(45)     NULL DEFAULT NULL,
  PRIMARY KEY (`export_id`),
  INDEX `idx_ea_actor` (`actor_employee_id`, `occurred_at` DESC),
  INDEX `idx_ea_type`  (`export_type`, `occurred_at` DESC)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
```

10. CLUSTER 12 — Lookups — Reconciliation

```
sql

-- =====
-- CLUSTER 12: LOOKUPS
-- =====

-- Strategy: cmms_parameter_master already has the right shape – extend.
-- It's a key-value lookup with (CategoryID, CategoryDescription, SrID, Value).
-- We add a real PK and audit columns.

ALTER TABLE `cmms_parameter_master`
  ADD COLUMN `is_active`    TINYINT(1) NOT NULL DEFAULT 1,
  ADD COLUMN `display_order` SMALLINT NOT NULL DEFAULT 0,
  ADD COLUMN `created_at`   DATETIME(6) NOT NULL DEFAULT CURRENT_TIMESTAMP(6),
  ADD COLUMN `updated_at`   DATETIME(6) NOT NULL DEFAULT CURRENT_TIMESTAMP(6)
                        ON UPDATE CURRENT_TIMESTAMP(6),
  ADD PRIMARY KEY (`CategoryID`, `SrID`),
  ADD INDEX `idx_pm_category` (`CategoryID`, `display_order`);

-- Isolate the backup variants so the runtime never accidentally reads them
RENAME TABLE
  `cmms_parameter_master_bkp`      TO `_legacy_parameter_master_bkp`,
  `cmms_parameter_master_jun2016` TO `_legacy_parameter_master_jun2016`,
  `cmms_parameter_master_incharge` TO `_legacy_parameter_master_incharge`;
```

MVP lookup seeds we will need (rows go in `cmms_parameter_master`):

- `CategoryID=1, 'Job Request Status'` → DRAFT, SUBMITTED, ASSIGNED, IN_PROGRESS, COMPLETED, VERIFIED_CLOSED, REJECTED, REOPENED
- `CategoryID=2, 'Equipment MVP Status'` → PENDING_VERIFICATION, ACTIVE, UNDER_CALIBRATION, ...
- `CategoryID=3, 'Calibration Status'` → VALID, DUE_SOON, OVERDUE, OUT_OF_TOLERANCE, NOT_REQUIRED
- `CategoryID=4, 'Priority'` → LOW, NORMAL, HIGH, URGENT
- `CategoryID=5, 'Equipment Category'` → T&ME, F&PE

(Note: enums on the actual `_MVP_STATUS` columns are the source of truth at the DB layer. Lookups are for UI display labels, ordering, i18n. Both exist by design.)

11. P2 CLUSTERS — Sketch Only

Per FINAL-DESC §25 Step 6, we sketch but do not finalise Phase 2 clusters.

Cluster	Existing tables	NEW for P2
5 Calibration	<code>cmms_jobcard_cal_*</code> , <code>cmms_ins_accuracy_info</code>	<code>calibration_certificates</code> , <code>standards</code> , <code>traceability_chain</code>
6 Maintenance	<code>cmms_amc_mst</code> , <code>cmms_device_spares_mst</code>	<code>pm_plans</code> , <code>breakdowns</code>
7 Scheduling	<code>cmms_schedule_mst</code> , <code>cmms_schedule equip_dtl</code>	<code>schedule_events</code> , <code>schedule_recurrence</code>
8 Procurement	<code>cmms_po_mst</code> , <code>cmms_pur_mst</code> , <code>cmms_inv_mst</code>	<code>vendors</code> (replaces missing <code>cmms_cont_mst</code> ?), <code>vendor_categories</code> , <code>stock_movements</code>
9 Documents	(none kept)	<code>documents</code> , <code>doc_versions</code> , <code>signatures</code>
11 Notifications	(none)	<code>notifications</code> , <code>notification_preferences</code>
13 Reporting	(none)	<code>saved_reports</code> , <code>report_runs</code>

These are **decision-needed** items I cannot resolve from the dump alone.

Q1 (● BLOCKING for Cluster 3) — Where is `cmms_cont_mst`?

4 tables reference `cmms_cont_mst (CMM_CONT_ID)` but its DDL is missing from the dump. This is the vendor/manufacturer master and is FK'd from `cmms equip_mst.EQM_MFRID` (5,704 rows depend on it). Can you export this table's schema?

Q2 (● Cluster 1) — Are SA79900 and AC77777 existing employees?

The 2 super-admin IDs you provided are 7 chars (fit `EMM_ID VARCHAR(7)`). Do they already exist in `cmms_emp_mst`'s 57 employees? If NO, the bootstrap migration needs to insert them first.

Please run:

```
sql
SELECT EMM_ID, EMM_NAME, EMM_INACTIVE FROM cmms_emp_mst WHERE EMM_ID IN ('SA79900', 'AC77777');
```

Q3 (● Cluster 1) — Migration of 565 existing user-role rows

For the 565 rows in `cmms_userrole_mst`:

- (a) Migrate them all and force password reset on first login? (RECOMMENDED — defence-grade)
- (b) Only migrate active employees (`cmms_emp_mst.EMM_INACTIVE = 0`)?
- (c) Skip migration entirely; let Super Admins provision users on demand?

Q4 (● Cluster 1) — Mapping 23 old roles → 5 new roles

The old `cmms_role_mst` has 23 role rows. We need a mapping. Can you share the names of the 23 roles? I'll propose a mapping table like:

```
old_role_id → new_role_code
1 ("Administrator") → SUPER_ADMIN
2 ("Lab In-Charge") → LAB_IN_CHARGE
...
23 ("Guest") → VIEW_ONLY
```

Q5 (● Cluster 3) — Can `cmms_division_hist` serve as `equipment_status_history`?

`cmms_division_hist` (3,676 rows) already tracks (EQM_TYPE, EQM_ID, EQD_STATUS, EQD_DIV_DATE). Should we extend it with `from_status` + `transitioned_by` instead of creating a new `equipment_status_history`? My preference: keep them separate (division history vs status history are semantically different).

Q6 (● Cluster 4) — Should we add `job_request_status_history`?

`cmms_jobcard_status_hist` (22,214 rows) only tracks job cards. Job requests have no status history table. Two options:

- (a) Create `job_request_status_history` (NEW, my recommendation)
- (b) Extend `cmms_jobcard_status_hist` with a polymorphic `entity_type` column (uglier but fewer tables)

Q7 (● Cluster 12) — Password policy strength?

For an ISRO SAC-like environment, what password policy should we enforce?

- Length minimum (8/12/14)?
- Complexity (mixed case + digit + symbol)?
- Expiry (90 days / 180 days / never)?
- Password history (last 3 / 5 / 10 forbidden)?

I included `password_history` in the DDL as optional. Confirm or remove.

Q8 (● Org context) — `cmms_emp_mst.EMM_DEPT` semantics

`EMM_DEPT` is an int FK to `cmms_section_mst(SM_ID)` (293 sections). Is "section" the right granularity for org-level row scoping (BR-VIS)? Or should we add a coarser "division" level above section?

13. PHASE 3 — DAY 2 PLAN

When DS returns answers to Open Questions 1-8, we complete:

1. Resolve `cmms_cont_mst` (Q1) → finalise Equipment cluster
 2. Verify SA79900/AC77777 (Q2) → finalise bootstrap seed
 3. Lock migration strategy for old users (Q3) + role mapping (Q4)
 4. Write final seed files:
 - `seed/01_roles.sql` (5 rows)
 - `seed/02_permissions.sql` (~45 rows mirroring the matrix)
 - `seed/03_role_permissions.sql` (full grant matrix per §6 FINAL-DESC)
 - `seed/04_super_admin_users.sql` (SA79900, AC77777)
 - `seed/05_lookup_values.sql` (statuses, frequencies, etc.)
 5. Draw clean ER diagrams for MVP-critical clusters (image deliverable)
 6. Set up the migration runner (versioned SQL files, naming convention)
 7. Lock the final MVP-critical schema and freeze
-

END OF PHASE 3 — DAY 1 OUTPUT

This document is subordinate to FINAL-DESC-CMCMIS v1.0. Any conflict, FINAL-DESC wins. Any decision in this document can be revised by DS — flag a "REVISION-CANDIDATE" and we update.